INCH-POUND
MIL-DTL-0050872D (AR)
29 March 2004
USED IN LIEU OF
MIL-C-50872C(AR)
22 July 1981
MIL-C-48082B (AR)
22 July 1981

DETAIL SPECIFICATION

CARTRIDGE, 40MM: HEDP, M433

This specification is approved for interim use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC). Other activities in the Department of Defense may use this interim revision or may continue using MIL-C-50872C (AR) and MIL-C-48082B (AR).

1. SCOPE

1.1 Scope. This specification covers the metal parts, assemblies, and the loading, assembling, and packing for one type of 40MM: High-Explosive, Dual-Purpose (HEDP) cartridge, designated Cartridge, 40MM: HEDP, M433.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806-5000 or e-mailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at http://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-5541	Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coating for Aluminum and Aluminum Alloys
MIL-P-20444	Primer, Percussion, M42, Parts for and Loading, Assembling, and Packaging
MIL-DTL-50869	Fuze PIBD: M550, Less Spitback Assembly
MIL-A-70625	Automated Acceptance Inspection Equipment Design, Testing and Approval, of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	DoD Preferred Method for Acceptance of Product
MIL-STD-2073-1	Standard Practice for Military Packaging

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

U.S. ARMY (ARDEC) DRAWINGS (see 6.3)

PRODUCT AND PACKING DRAWINGS

8835104 - Box, Wirebound, Packing, Ammunition for Cartridge 40MM
8835105 - Box, Fiber, Packing, Ammunition for Cartridge 40MM
8844609 - Cartridge Case Loading Assembly M118
8844610 - Case, Cartridge 40MM: M118
8844611 - Plug, Base
8844612 - Cup, Powder Charge
8862362 - Box, Fiber, Packing, Ammunition for Cartridge, 40MM: M397, M397E1 and M433
8882363 - Box, Wirebound, Packing, Ammunition for Cartridge, 40MM: M397, M397E1, M433 and M407A2
8886340 - Ring, Skirt
8886344 - Skirt
8886349 - Fuze, PIBD: M550
8886351 - Spitback Assembly

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8886352 - Liner Spitback
8886353 - Spitback, Loading Assembly
8886354 - Housing Spitback
8886360 - Fuze, PIBD: M550, Less Spitback
8886371 - Cartridge, 40MM: HEPD, M433
8886372 - Projectile Assembly for Cartridge, 40MM: HEPD, M433
8886374 - Cup & Liner Loading Assembly
8886375 - Cup and Skirt Assembly
9202576 - Tensile Specimen
9276544 - Liner (One Piece)
9276546 - Cup, Boattail

INSPECTION EQUIPMENT DRAWINGS

8827889 - Gage, Receiver, Profile and Alignment, 40MM Cartridge
8827890 - Gage, Plug, Acceptance Check
8870392 - Ultrasonic Test Equipment
8827895 - Limit Check

(Copies of these drawings may be requested online at drawing-request@pica.army.mil, or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of the documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS INTERNATIONAL (ASTM)

ASTM E8	Standard Test Methods for Tension
ASTM E18	Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM B117	Standard Method of Salt Spray (Fog) Testing
ASTM E1255	Standard Practice for Radioscopy

(Copies may be requested online at www.astm.org or ordered from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the solicitation or contract order (see 6.2), a sample shall be subjected to first article in accordance with 4.3.

3.2 Materials. Materials, parts and assemblies shall comply with requirements specified on the applicable drawings and referenced specifications.

3.3 Fuze safety. The fuze shall not be armed or partially armed. The fuze shall comply with all requirements of MIL-DTL-50869.

3.4 Primer functioning. The primer shall function without evidence of squibbing or hang fire (see MIL-P-20444 for definitions). There shall be no primer blow back as evidenced by perforation of the cup and extrusion of the primer around the firing pin causing difficulty in opening weapon.

3.5 Projectile velocity. The projectiles shall have a mean velocity of 247 ± 7 feet per second (fps) and a standard deviation not exceeding 4.5 fps (see 6.4.1).

3.6 Projectile functioning. The projectiles shall function high order as evidenced by fragmentation of the body as well as being capable of defeating 2 ½ inch mild steel targets.

3.7 Safety. There shall be no projectile premature burst (see 6.4.3) or early burst (6.4.2) in any of the test firings.

3.8 Extraction. The cartridge case shall be ejected from the launcher by a single forward slide of the barrel.

3.9 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.10 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough and workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

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4. VERIFICATION

TABLE I. Requirement/Inspection Cross Reference Matrix

Method of inspection
N/A – Not Applicable (N/A)
1 – Analysis
2 – Demonstration
3 – Examination
4 – Test

Classes of inspection
A – First article
B – Conformance

SECTION 3 REQUIREMENT	INSPECTION METHODS					INSPECTION CLASS		SECTION 4 INSPECTION METHOD
	N/A	1	2	3	4	A	B	
3.1			X	X	X	X		4.3
3.2			X	X	X	X	X	4.4
3.3				X		X	X	4.5.18
3.4					X	X	X	4.5.9, 4.5.11, 4.5.23
3.5				X	X	X	X	4.5.23
3.6				X	X	X	X	4.5.23
3.7				X	X	X	X	4.5.23
3.8				X		X	X	4.5.23
3.9				X		X	X	4.5.25
3.10				X		X	X	4.5.28

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (4.3)
b. Conformance inspection (4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in section 4.5.

4.3 First article inspection. When specified in the solicitation or contract, a sample of the cartridges shall be subjected to first article inspection.

4.3.1 First article quantity. The contractor shall submit a first article for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Parts description	Drawings	Quantity
Cartridge Case Loading Assy. M118	8844609	410[1]
Cartridge Case	8844610	100
Base Plug	8844611	50
Cup, Powder Charge	8844612	50
Projectile Assembly	8886372	410[1]
Fuze Assembly	8886349	410[1]

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Fuze	8886360	410[1]
Spitback Assembly	8886351	25
Liner Spitback	8886352	25
Spitback Assy	8886353	25
Housing Spitback	8886354	25
Cup and Liner Loading Assy.	8886374	410[1]
Cup and Skirt Assembly	8886375	50
Skirt	8886344	50
Cup, Boattail	9276546	50
- (Embossed stock prior to forming cup)		2[2]
Liner	9276544	15
Retaining Ring	8886340	25
- (Prior to heat treatment)		25
Cartridge	8886371	410[1]

[1] - All quantities shall be subcomponents within final assembly. The subcomponents shall be individually examined for all listed characteristics prior to further assembly.
[2] - Size of sample - width of coil by 20 inches.

4.3.2 Inspections to be performed. The first article inspection shall be 100% examination for defects of all drawing characteristics (listed and unlisted) including workmanship requirements for the parts and assemblies listed in 4.3.1 and the tests specified in Table II herein.

4.3.3 First article rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE II. First Article Inspection.

Cartridge Assembly and Components

EXAMINATION OR TEST	NO. OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Cartridge Case (Dwg. 8844610)			
Hardness	100	3.2	4.5.2
Salt spray	10	3.2	4.5.4
Diameter of vent holes (Sectioned)	25	3.2	Gage/4.5.7
Hydrostatic 2/	100	3.2	4.5.5
Tensile	6	3.2	4.5.6
Base Plug (Dwg. 8844611)			
Salt spray	10	3.2	4.5.4
Cartridge Case Loading Assembly (Dwg. 8844609)			
Security of crimp of base	20	3.2	4.5.8
Skirt (Dwg. 8886344)			
Salt spray	4	3.2	4.5.4
Cup and Skirt Assembly (8886375)			
Air pressure	50	3.2	4.5.14
Push out	50	3.2	4.5.15
Retaining Ring (Dwg. 8886340)			
Push out	25	3.2	4.5.16
Fuze, PIBD, M550 (Dwg. 8886349)			
X-Ray	360	3.3	4.5.18
Cartridge (Dwg. 8886371)			
Functioning	288	3.4 to 3.8	4.5.23
Environmental	72	1/	4.5.24
Air pressure	50	3.2	4.5.22
Pull Test	50	3.2	4.5.21

NOTES:

1/ These tests are for informational purposes only.

2/ Cases especially prepared for this test shall be identified from regular cases by painting a red strip around the open end, or any positive identification approved by the Contracting Officer.

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4.4 Conformance inspection.

4.4.1 Inspection lot formation. Lot formation shall be in accordance with MIL-STD-1916. For the components identified below, each inspection lot of cartridges shall be produced with components complying with the following constraints:

a. M118 cartridge case loading assemblies metal parts from one interfix number, from one manufacturer.
b. Projectile metal parts from one lot interfix number, from one manufacturer.
c. Fuzes from no more than two consecutively produced serial lots from one manufacturer.
d. Primers from one lot number from one manufacturer.
e. Liners and caps from one interfix lot number from one manufacturer.
f. RDX or Composition A-5 from one lot interfix number, from one manufacturer.
g. Loaded spitback assemblies from one lot interfix number, from one manufacturer.
h. Propellant from not more than one lot number from one manufacturer.

4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for Critical, Major and Minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Verification Level IV shall be used for all characteristics defined as Majors and Verification Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.4.2.1 through 4.4.2.23. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

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Conformance Inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.1	Cartridge Case	SHEET 1 OF 2	8844610
			NEXT HIGHER ASSEMBLY 8844609

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1.	Three (3) or more vent holes blocked or missing	100%	3.2	AAIE/4.5.1
MAJOR				
101.	One or two vent holes blocked or missing	100%	3.2	AAIE/4.5.1
102.	Hardness	Level IV	3.2	4.5.2
103.	Process control inspection of bath temperature for anodic coating	3/	3.2	4.5.3
104.	Salt spray	1/	3.2	4.5.4
105.	Diameter of vent holes	Level IV	3.2	Gage or 4.5.7
106.	True position of largest outside diameter with second largest diameter	Level IV	3.2	Gage
107.	True position of largest inside diameter with second largest diameter	Level IV	3.2	Gage
108.	True position of base plug cavity with second largest outside diameter	Level IV	3.2	Gage
109.	True position of powder charge cavity with base plug cavity	Level IV	3.2	Gage
110.	Diameter of powder charge cavity, min	Level IV	3.2	Gage
111.	Diameter of base plug cavity	Level IV	3.2	Gage
112.	Depth of base plug cavity	Level IV	3.2	Gage
113.	Depth from base to bottom of powder charge cup cavity	Level IV	3.2	Gage
114.	Second largest outside diameter	Level IV	3.2	Gage 2/
115.	Diameter of node	Level IV	3.2	Gage
116.	Parallelism of open end to base	Level IV	3.2	Gage
117.	Perpendicularity of base to second largest outside diameter	Level IV	3.2	Gage
118.	Largest inside diameter	Level IV	3.2	Gage 4/
119.	Distance of flat of inside projection	Level IV	3.2	Gage
120.	Diameter of flange, max	Level IV	3.2	Gage
121.	Total length	Level IV	3.2	Gage
122.	Diameter at datum length, min	Level IV	3.2	Gage

NOTES:

1/ Sampling and rejection shall be in accordance with MIL-A-8625.

2/ If the requirement cannot be met in the free state, to “go” inside diameter plug gage may be inserted in the cartridge case while inspecting the “go” and “no-go” for this characteristic. If used, the data shall be annotated to note use of the “go” inside diameter plug gage.

3/ Continuous monitoring inspection.

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Conformance Inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.1	Cartridge Case (con't)	SHEET 2 OF 2	8844610
			NEXT HIGHER ASSEMBLY 8844609

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
MAJOR				
123.	Unconfined second largest outside diameter	Level IV	3.2	Gage 5/
124.	Burr in vent hole	Level IV	3.2	Gage
125.	Metal defective 6/	Level IV	3.9	Visual
126.	Tensile properties	7/	3.2	4.5.6
MINOR				
201.	Thickness of flange	Level II	3.2	Gage
202.	Length of base plug seating projected from bottom of base	Level II	3.2	Gage
203.	Wall thickness, min	Level II	3.2	Gage
204.	Stamping misleading or unidentifiable	Level II	3.2	Visual
205.	Protective coating with bare spot except anode rack marks	Level II	3.2	Visual
206.	Color incorrect	Level II	3.2	Visual
207.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:

4/ If the requirement cannot be met in the free state, a confining fixture may be utilized while inspecting the "go" and "no-go" for this characteristic. If used, the data shall be annotated to note use of the fixture on outside diameter.

5/ Case must fall freely without any force into the applicable gage.

6/ Nick, crack, dent or sharp edges which might interfere with the proper assembly or end item performance.

7/ Six (6) specimens to be taken from each heat treat batch. Accept on zero (0) reject on one (1).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.2	Base Plug	SHEET 1 OF 1	8844611
			NEXT HIGHER ASSEMBLY 8844609

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1.	Depth of primer cavity	100%	3.2	AAIE/4.5.1
MAJOR				
101.	Process control inspection of bath temperature for anodic coating	2/	3.2	4.5.3
102.	Salt spray	1/	3.2	4.5.4
103.	Largest outside diameter	Level IV	3.2	Gage
104.	True position of smallest outside diameter with largest outside diameter	Level IV	3.2	Gage
105.	True position of primer cavity diameter with largest outside diameter	Level IV	3.2	Gage
106.	Smallest outside diameter	Level IV	3.2	Gage
107.	Diameter of primer cavity	Level IV	3.2	Gage
108.	Length of primer end to shoulder including angle	Level IV	3.2	Gage
109.	Total length	Level IV	3.2	Gage
110.	Diameter of flash hole	Level IV	3.2	Gage
111.	Radius of intersection of primer cavity missing	Level IV	3.2	Visual
MINOR				
201.	Second largest outside diameter	Level II	3.2	Gage
202.	Length from primer end to far side of shoulder	Level II	3.2	Gage
203.	Protective coating with bare spot except anode rack marks	Level II	3.2	Visual
204.	Finish improper	Level II	3.2	Visual
205.	Color incorrect	Level II	3.2	Visual
206.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:
1/ Sampling and rejection shall be in accordance with MIL-A-8625.
2/ Continuous monitoring inspection.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.3	Powder Charge Cup	SHEET 1 OF 1	8844612
			NEXT HIGHER ASSEMBLY 8844609

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Length to datum	Level IV	3.2	Gage
102.	Inside diameter	Level IV	3.2	Gage
103.	Diameter of flange, maximum	Level IV	3.2	Gage
MINOR				
201.	Wall thickness	Level II	3.2	Gage
202.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28
NOTES:				

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.4	Cartridge Case Loading Assembly Prior to Assembling Base Plug and Primer	SHEET 1 OF 1	8844609
			NEXT HIGHER ASSEMBLY 8886371

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL 1.	Propellant charge weight more than 25% above or below nominal load weight	100% 1/	3.2	AAIE/4.5.1/4.5.27
MAJOR 101.	Propellant charge weight outside tolerance range but less than 25% of nominal	100% 1/	3.2	AAIE/4.5.1/4.5.27
MINOR 201.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:
1/ Any cartridge found to have a propellant weight out of tolerance but within 25% of the nominal load weight shall be removed from the lot.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.5	Cartridge Case Loading Assembly	SHEET 1 OF 1	8844609
			NEXT HIGHER ASSEMBLY 8886371

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1.	Powder charge cup missing	100%	3.2	AAIE/4.5.1 1/
2.	Primer above flush with respect to base of case	100%	3.2	AAIE/4.5.1
MAJOR				
101.	Security of crimp of base plug	32 6/	3.2	4.5.8 2/
102.	Base plug above flush	Level IV	3.2	Gage
103.	Depth to base plug, maximum	Level IV	3.2	Gage
104.	Primer above flush or more than .005 below with respect to base plug	Level IV	3.2	Gage
105.	Primer function	800	3.2	4.5.9
106.	Primer deterioration	3/	3.2	4.5.10
107.	Production control check on loaded cases	4/	3.4	4.5.11
MINOR				
201.	Excessive varnish on exterior surface of primer	Level II	3.2	Visual
202.	Marking misleading or unidentifiable 5/	Level II	3.2	Visual
203.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:

1/ Inspection may occur prior to assembling base plug and primer.
2/ Special build units without propellant load.
3/ Sampling and rejection shall be in accordance with MIL-P-20444.
4/ Ten (10) randomly selected from every 4 hours production from each primer loading machine.
5/ See Dwg. 8844610 for marking requirements.
6/ Accept on zero (0) and reject on (1).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.6	Liner Spitback	SHEET 1 OF 1	8886352
			NEXT HIGHER ASSEMBLY 8886351

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Outside diameter	Level IV	3.2	Gage
102.	Minimum wall thickness, after forming	Level IV	3.2	Gage
MINOR				
201.	Spherical radius	Level II	3.2	Gage
202.	Height	Level II	3.2	Gage
203.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28
NOTES:				

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.7	Housing Spitback	SHEET 1 OF 1	8886354
			NEXT HIGHER ASSEMBLY 8886353

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None defined			
MAJOR				
101.	Smallest outside diameter	Level IV	3.2	Gage
102.	Second largest inside diameter	Level IV	3.2	Gage
103.	True position of second largest inside diameter with smallest outside diameter including perpendicularity with flange.	Level IV	3.2	Gage
104.	Diameter of counterbore	Level IV	3.2	Gage
105.	True position of counterbore with second largest inside diameter	Level IV	3.2	Gage
106.	Distance from small end to flange	Level IV	3.2	Gage
107.	Depth of counterbore	Level IV	3.2	Gage
108.	Location small basic diameter	Level IV	3.2	Gage
109.	Location large basic diameter	Level IV	3.2	Gage
MINOR				
201.	Distance from large end to front face of flange	Level II	3.2	Gage
202.	Thickness of flange	Level II	3.2	Gage
203.	Diameter of flange	Level II	3.2	Gage
204.	Large chamfer on flange missing	Level II	3.2	Visual
205.	Body diameter	Level II	3.2	Gage
206.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:

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Conformance Inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.8	Spitback Loading Assembly	SHEET 1 OF 1	8886353
			NEXT HIGHER ASSEMBLY 8886351

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Depth of spherical cavity	Level IV	3.2	Gage
MINOR				
201.	Caked explosive on liner assembly	Level II	3.2	Visual
202.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28
NOTES:				

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.9	Spitback Assembly	SHEET 1 OF 1	8886351
			NEXT HIGHER ASSEMBLY 8886349

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Length from forward flange to crimped end	Level IV	3.2	Gage
102.	Swell diameter	Level IV	3.2	Gage
103.	Depth to charge, small end	Level IV	3.2	Gage
104.	Charge above flush, small end	Level IV	3.2	Gage
MINOR				
201.	Crimp not full 360°	Level II	3.2	Visual
202.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.10	Liner (One Piece)	SHEET 1 OF 1	9276544
			NEXT HIGHER ASSEMBLY 8886374

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Diameter of flange	Level IV	3.2	Gage
102.	Wall thickness	Level IV	3.2	Gage
103.	Location large basic diameter	Level IV	3.2	Gage
104.	Location small basic diameter	Level IV	3.2	Gage
105.	True position of hole small end with flange	Level IV	3.2	Gage
106.	Runout of small end with flange diameter	Level IV	3.2	Gage
MINOR				
201.	Length from small end to flange	Level II	3.2	Gage
202.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.11	Skirt	SHEET 1 OF 1	8886344-1
			NEXT HIGHER ASSEMBLY 8886375

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Distance between fuze seating surface and cup seating surface	Level IV	3.2	Gage
102.	True position of rotating bands with respect to bourrelet diameter	Level IV	3.2	Gage
103.	Diameter rotating band	Level IV	3.2	Gage
104.	Salt spray test	1/	3.2	4.5.4
105.	Diameter of fuze cavity	Level IV	3.2	Gage
106.	True position of fuze cavity diameter with respect to bourrelet diameter	Level IV	3.2	Gage
107.	True position of cup cavity diameter with respect to bourrelet diameter.	Level IV	3.2	Gage
108.	Incorrect color	Level IV	3.2	Visual
MINOR				
201.	Total length	Level II	3.2	Gage
202.	Diameter rear of rotating bands	Level II	3.2	Gage
203.	Diameter of cup cavity	Level II	3.2	Gage
204.	Smallest outside diameter	Level II	3.2	Gage
205.	Diameter between rotating bands	Level II	3.2	Gage
206.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:
1/ Sampling and rejection shall be in accordance with MIL-C-5541.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.12	Boattail Cup	SHEET 1 OF 1	9276546
			NEXT HIGHER ASSEMBLY 8886375

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Presence of cracks, voids, pipes or inclusions, etc.	100%	3.2	4.5.12
102.	Outside diameter, maximum	Level IV	3.2	Gage
103.	Total length	Level IV	3.2	Gage
MINOR				
201.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.13	Cup and Skirt Assembly	SHEET 1 OF 1	8886375
			NEXT HIGHER ASSEMBLY 8886374

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1.	Cracked cup 2/	100%	3.2	AAIE/4.5.1
MAJOR				
101.	Bourrelet diameter	Level IV	3.2	Gage
102.	Rotating band diameter, maximum	Level IV	3.2	Gage
103.	Runout of boattail surface with fuze cavity depth	Level IV	3.2	Gage
104.	Adhesive not 360°	Level IV	3.2	Visual
105.	Length skirt to rear rotating band, maximum	Level IV	3.2	Gage
106.	Air pressure test	100%	3.2	4.5.14
107.	Pushout test	80 1/	3.2	4.5.15
MINOR				
201.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:
1/ Accept on zero (0) and reject one (1).
2/ Examination for cracked cups shall occur after air pressure test.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.14	Retaining Ring	SHEET 1 OF 1	8886340
			NEXT HIGHER ASSEMBLY 8886374

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Inside diameter 1/	Level IV	3.2	Gage
102.	Outside diameter 1/	Level IV	3.2	Gage
103.	True position of inside diameter with outside diameter 1/	Level IV	3.2	Gage
104.	Push out test	80 2/	3.2	4.5.16
MINOR				
201.	Total height 1/	Level II	3.2	Gage
202.	Protective finish with bare spot	Level II	3.2	Visual
203.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:
1/ Prior to heat treatment.
2/ Accept on zero (0) and reject one (1).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.15	Cup and Liner Loading Assembly	SHEET 1 OF 1	8886374
			NEXT HIGHER ASSEMBLY 8886372

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1.	Cracked cup	100%	3.2	AAIE/4.5.1
MAJOR				
101.	Liner missing or loose	100%	3.2	Visual
102.	Retaining ring missing	100%	3.2	Visual
103.	Distance from fuze seating surface to top of liner	Level IV	3.2	Gage
103.	Diameter of rotating band	Level IV	3.2	Gage
105.	True position of bottom diameter of liner with inside diameter of skirt	Level IV	3.2	Gage
106.	Cap missing, loose or torn exposing explosive	100%	3.2	Visual
107.	Specific gravity of charge	80 1/	3.2	4.5.17
MINOR				
201.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:
1/ Accept on zero (0) and reject on one (1).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.16	Fuze, PIBD, M550, Less Spitback	SHEET 1 OF 1	8886360
			NEXT HIGHER ASSEMBLY 8886349

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL				
1.	X-ray examination 1/	100%	3.3	4.5.18
MAJOR				
101.	X-ray examination 2/	100%	3.3	4.5.18
MINOR				
201.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:

1/ Critical Defect is defined as:

a. Setback pin not seated in bottom of counterbore of escapement housing or retracted.
b. Setback pin missing.
c. Setback pin spring missing.
d. Detent missing.
e. Detent spring missing.
f. Detent end not in contact or seated within root of teeth of rotor gear.
g. Pinion assembly missing.
h. Verge assembly missing.
i. Rotor gear asssembly in armed, or partially armed, position.
j. Setback pin not blocking rotor.

2/ Major Defect is defined as:

a. Foreign material present.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.17	Fuze, PIBD: M550	SHEET 1 OF 1	8886349
			NEXT HIGHER ASSEMBLY 8886372

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	True position of outside diameter of spitback with fuze outside diameter including perpendicularity with fuze seating surface	Level IV	3.2	Gage
102.	Staking missing or improper	Level IV	3.2	Visual
103.	Pull test of spitback assembly	120 1/	3.2	4.5.19
MINOR				
201.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:
1/ If any spitbacks separate from the fuze (partially or completely) the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.18	Projectile Assembly	SHEET 1 OF 1	8886372
			NEXT HIGHER ASSEMBLY 8886371

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Weight incorrect	Level IV	3.2	Gage
102.	Torque of crimp of fuze assembly	Level III	3.2	4.5.20
MINOR				
201.	Length from front end to rear of rotating bands	Level II	3.2	Gage
202.	Crimp improper	Level II	3.2	Visual
203.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.19	Cartridge	SHEET 1 OF 1	8886371
			NEXT HIGHER ASSEMBLY ----------

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Projectile pull test	Level III	3.2	4.5.21
102.	Air pressure	Level III	3.2	4.5.22
103.	Chamber gage failure (after complete cure)	100%	3.2	8827889, 8827890, 8827895
104.	Function	288	3.4 to 3.8	4.5.23
105.	Environmental	72 1/	2/	4.5.24
MINOR				
201.	Total length	Level II	3.2	Gage
202.	Marking misleading or unidentifiable	Level II	3.2	Visual
203.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:

1/ These tests (4.5.24 a., b., c.) shall be conducted until three consecutive lots have successfully met all Function requirements for tests defined by Major 104 at which time only 4.5.24 c. shall be required for LAT.

2/ These tests are for informational purposes only.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.20	Unsealed fiber box	SHEET 1 OF 1	8882362 or 8835105
			NEXT HIGHER ASSEMBLY ----------

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Incorrect number of cartridges	Level IV	3.2	Visual
102.	Separator missing	Level IV	3.2	Visual
MINOR				
201.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28
NOTES:				

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.21	Sealed fiber box, prior to sealing bag	SHEET 1 OF 1	8882362 or 8835105
			NEXT HIGHER ASSEMBLY ----------

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Heat seal test of packing seals	80 1/	3.2	4.5.26
102.	Contained damaged	Level IV	3.2	Visual
103.	Corners not blunted	Level IV	3.2	Visual
MINOR				
201.	Contents loose	Level II	3.2	Visual
202.	Marking misleading or unidentifiable	Level II	3.2	Visual/4.5.28
203.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:
1/ Accept on zero (0) and reject on one (1).

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.22	Unsealed wooden packing box	SHEET 1 OF 1	8882363 or 8835104
			NEXT HIGHER ASSEMBLY ----------

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Bag torn or perforated	Level IV	3.2	Visual
102.	Bag improperly sealed	Level IV	3.2	Visual
103.	Marking of bag misleading or unidentifiable	Level IV	3.2	Visual
MINOR				
201.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.23	Sealed wooden packing box	SHEET 1 OF 1	8882363 or 8835104
			NEXT HIGHER ASSEMBLY ----------

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None Defined			
MAJOR				
101.	Box damaged	Level IV	3.2	Visual
MINOR				
201.	Contents loose	Level II	3.2	Manual
202.	Car seal missing or improperly positioned	Level II	3.2	Visual
203.	Marking misleading or unidentifiable	Level II	3.2	Visual
204.	Evidence of poor workmanship	Level II	3.10	Visual/4.5.28

NOTES:

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4.5 Methods of inspection

4.5.1 Inspection Equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the Inspection Method Column in the tables starting with paragraph 4.4.2.1. Inspection equipment used for acceptance of product shall be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% or better of the total tolerance of the characteristic being inspected. Any automated acceptance inspection equipment (AAIE) prove-out shall be performed in accordance with MIL-A-70625 (see 6.5).

4.5.2 Hardness. The cartridge case shall be subjected to the Rockwell hardness test in accordance with ASTM E18. This test is a non-destructive test. Parts so tested may be returned to the lot. The cartridge case hardness requirement is specified in 3.2.

4.5.3 Process control inspection of bath temperature for anodic coating. Continuous monitoring inspection shall be conducted for bath temperature requirement specified on the applicable drawings. If the bath temperature for a batch of parts does not comply with the applicable drawing requirement, the batch shall be suitably identified, classified defective and removed from the lot.

4.5.4 Salt spray. The specified part shall be subjected to the salt spray test in accordance with ASTM B 117. This test shall be conducted for 72 hours, or for the skirt (Dwg. 8886344) the duration specified on the applicable drawing, using a salt solution of 5%. This test is a destructive test. Parts so tested shall not be returned to the lot. Sampling and rejection criteria is as specified in accordance with MIL-A-8625.

4.5.5 Hydrostatic pressure of cartridge case. The cartridge case, less vent holes, shall be placed in a Government-approved test fixture. The protrusions inside the case shall not be supported. The fixture shall be assembled in a hydrostatic machine and a load applied, no greater than 0.25 inch per minute, until the required pressure of 48000 pounds per square inch (psi), minimum(min) is reached. This is a destructive test, and parts so tested shall not be returned to the lot.

4.5.6 Tensile strength of cartridge case. The sample of the tensile strength specimen shall be as specified on (Dwg. 9202576). The specimen shall be taken from each heat-treat batch prior to trimming and anodizing from a ¾ inch wide hoop, ½ inch in from the mouth of the cartridge case. The hoop shall be split, unrolled and cut to the required shape. The test shall be conducted in accordance with ASTM E8 and shall comply with the requirements of the applicable drawing.

4.5.7 Total vent hole area (alternate method). In lieu of measuring vent hole diameters the cartridges may be inspected for total vent hole area The cartridge case shall be placed in a Government-approved test fixture. The maximum limits of the air column shall be set with masters (one of each) having six holes with a max. total not greater than .0356 in^2 and a min. total area not less than .0317 in^2.

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4.5.8 Security of crimp of base plug on cartridge case assembly. The assembly shall be placed in an approved fixture and the minimum unseating force specified on the applicable drawing shall be applied. The force shall then be increased to the dead load required minimum of the applicable drawing. If the average unseating force for the sample is less than the required average or if any one sample has an unseating force below the minimum, on the applicable drawing, the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916.

4.5.9 Primer functioning. Sampling shall be in accordance with paragraph 4.2.3.2.2 of MIL-P-20444C, except that the sample size shall be 800 primers from each lot. Each primer of the 800 samples shall be tested as specified in MIL-P-20444C, paragraph 4.3.2.2. If one or more primers fail to comply with the applicable requirement, the lot shall be rejected.

4.5.10 Check test for deterioration of primers. If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass, the check test for deterioration, specified in MIL-P-20444, immediately before the primer lot is assembled into the cartridge. This test shall be performed by the contractor on primers selected by the Government Inspector at the facility assembling the primers into the cartridge.

4.5.11 Production control check on loaded M118 cartridge case. The loaded M118 cases produced by each primer assembly machine shall be kept segregated and identified. They shall not be used until this test has been performed. The loaded M118 cartridge case shall be fired from an approved launcher placed on an approved mount. No projectile is required for this test. Observation shall be made for primer functioning. If any sample fails to comply with the applicable requirements, the four (4) hours production represented by the sample shall be rejected. Parts so tested shall not be returned to the lot.

4.5.12 Ultrasonic inspection. Cup, boattail shall be inspected for the presence of cracks, voids, pipes, or inclusions, etc., along the cylindrical section of the cup throughout the entire wall thickness including base section, gaged externally to a depth as specified on the applicable drawings. This test shall be performed and calibrated as shown on the equipment drawing (Dwg. 8870392). Any component not meeting the requirements of Dwg. 8870392 shall be scrapped and not returned to the lot. If calibration shows equipment not to be performing correctly, the quantity represented prior to calibration shall be reprocessed.

4.5.13 Dimensional inspection of boattail cup, embossed stock (see Dwg. 9276546). Two samples of the embossed stock shall be selected; one sample shall be selected from the beginning of the embossed coil, and one sample shall be selected from the end of the embossed coil. The length of each sample shall be, as a minimum, equal to the circumference of the embossing roller (in order to ensure that the embossed samples represent the entire embossing roller). The width of each sample shall be 6 inches. Fifteen measurements shall be conducted on the sample strip for each characteristic. There shall be three points of measurements across the width of the strip at five points along the length, uniformly spaced. This process shall be repeated on sample specimens when each new, repaired, or adjusted embossing roller enters the

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production line. If the average of the 15 measurements for depth of depression and material thickness fails to comply with the drawing dimensions and tolerances, the embossed plate stock, from which the sample was taken, shall be rejected. Corrective action shall be performed prior to further production.

4.5.14 Air pressure test of cup and skirt assembly. The cup and skirt assembly shall be placed in an approved fixture and the minimum pressure for the minimum duration specified on the applicable drawing shall be applied. If any sample's joint between the cup and skirt fails during this test the lot represented by the sample shall be rejected and set aside for disposition in accordance with MIL-STD-1916. This test is considered a non-destructive test provided the pressure applied does not exceed 10% of that specified on the drawing and the duration of pressure application does not exceed 100% of the specified minimum time. Parts so tested may be returned to the lot.

4.5.15 Push out test of cup and skirt assembly. The cup and skirt assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. If any sample's joint between the cup and skirt fails during this test the lot represented by the sample shall be rejected and set aside for disposition in accordance with MIL-STD-1916. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.16 Push out test of retaining ring (after heat treatment). The retaining ring shall be placed in an approved fixture and the minimum push out force specified on the applicable drawing shall be applied at a rate of ½ in per minute. If any retaining rings fail to comply with the requirements specified on the applicable drawing, the lot represented by the sample shall be rejected and set aside for disposition in accordance with MIL-STD-1916. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.17 Specific gravity of the charge of cup and liner loading assembly. If any sample fails to comply with the specific gravity requirements, specified on the applicable drawing, the lot represented by the sample shall be rejected.

4.5.17.1 Method for specific gravity determination. The specific gravity of the charge may be determined by the following method:

a. Mark the cup and liner assembly for identification.
b. Weigh the empty cup and liner assembly and record as W_e.
c. Level cup and liner with a leveling device. Fill the cup and liner assembly with water, weigh and record as $W_{W.}$
d. Pour as much water as possible from cup and liner assembly, dry interior of skirt with a clean cotton cloth, or piece of absorbent paper, and place assembly, open end down, on absorbent paper. Allow assembly to dry.
e. Distribute preweighed assemblies, among other assemblies, and load in regular manner with RDX.
f. Retrieve the loading assembly and record as W_1.
g. Subtract (W_1 - W_e) to obtain the explosive weight.

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h. Level the assembly with leveling device. Fill assembly with water using care to fill to same point used when filling empty cup and liner with water and record as W_b.
i. Empty water from assembly and thoroughly dry surface of liner and assembly interior with a clean cloth or absorbent paper.
j. Calculate specific gravity as follows:

$$\text{Specific Gravity} = \frac{W_1 - W_e}{(W_W - W_e) - (W_b - W_1)}$$

4.5.17.2 Alternate method for specific gravity determination.

a. Prior to assembly, weigh empty body assembly with liner assembly and retainer in air (grams). Record the weight as A.
b. Prior to assembly, weigh the same empty body assembly with liner assembly and retainer submerged in water (grams). Record the weight as B.
c. Dry the components, and load with explosive composition using normal production methods.
d. Weigh the loaded assembly in air(grams). Record the weight as C.
e. Weigh the loaded assembly submerged in water (grams). Record the weight as D.
f. Calculate the specific gravity of the charge as follows:

$$\text{Specific Gravity} = \frac{C - A}{(C - A) - (D - B)}$$

4.5.18 X-ray examination of fuze less spitback. Radioscopic examination equipment, procedures, and personnel shall be in accordance with the requirements of ASTM E1255 Annex A1. Each fuze assembly prior to assembling to spitback assembly shall be x-rayed. The inspection shall be conducted in two planes; from the top of the fuze down, and from the side. Each view shall be read by two independent inspectors. If any fuze shows evidence of being improperly assembled, or having missing parts, it shall be rejected. If any fuze shows evidence of any of the critical defects listed in 4.4.2.16 (note 1), the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916.

4.5.19 Pull test of spitback assembly from fuze assembly. The fuze assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied. If any spitbacks separate from the fuze (partially or completely) the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916. This test is considered a nondestructive test provided the force applied does not exceed 10% of the minimum specified on the applicable drawing. Parts so tested may be returned to the lot.

WARNING: This test is conducted on assemblies containing an explosive element. Test barricades (where used), procedures and equipment shall have prior approval.

4.5.20 Torque of crimp on fuze assembly on projectile. The projectile shall be placed in an approved fixture and the minimum torque specified on the applicable drawing shall be

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applied. If any projectiles rotate under the minimum torque the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916. This is considered a nondestructive test provided the torque applied does not exceed 10% of the minimum specified on the applicable drawing. Parts so tested may be returned to the lot.

4.5.21 Projectile pull test. The cartridge shall be placed into a Government-approved pull-test fixture and the axial pull force necessary to separate the cartridge shall be measured. If for any cartridge the separation force is not within the limits specified by the applicable drawing the lot shall be rejected and set aside for disposition in accordance with MIL-STD-1916. The separated parts may be reworked, reassembled, and returned to the lot provided that it is done in accordance with a rework procedure approved by the Contracting Officer.

NOTE: This test is conducted on assemblies having explosive elements. Appropriate safety procedures must be in place during this test.

4.5.22 Air pressure test of cartridge. The cartridge shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. Observation shall be made for conformance of the requirement on the applicable drawing.

NOTE: Rounds in which it can be proven that the source of leaking is the ogive-to-fuze crimp are to be considered a no-test round and shall not be considered as a basis for rejecting a lot.

4.5.23 Functioning. The sample cartridges shall be randomly pulled from the lot for ballistic testing. The selection shall be by the Government inspector, or contractor if approved by the Contracting Officer. Any reworked cartridges shall be represented within this sample on a prorated basis. Procedures and rejection criteria are defined below.

4.5.23.1 Transportation vibration and X-ray examination. Prior to forwarding the cartridges to the Government approved facility for ballistic testing, they shall be subjected to x-ray examination using x-ray equipment or any other method satisfactory to the contracting officer. If any critical defect is found (per 4.5.18), the lot shall be rejected, including the ballistic samples. If any major defect is found, it shall be noted, but not replaced in the ballistic sample. Replacement of any ballistic sample shall be cause for rejecting the lot. The cartridges shall then be packed in their regular shipping containers and be tested in accordance with MIL-STD-331 (Test No.B2, with vibration levels conforming to 6.1.2.1 of Test B1). The test shall be performed at ambient temperature only. Simulated cartridges may be used to insure a full container. After completion of the transportation vibration test the cartridges shall be x-rayed again in accordance with 4.5.18. If one or more cartridge's fuzes fails to meet the requirements of 4.5.18 or has experienced damage that compromises any safety feature of the fuze the lot shall be rejected. Cartridges subjected to transportation vibration tests shall not be returned to the lot. Only those cartridges selected for functioning tests shall be subjected to this test.

4.5.23.2 Ballistic testing. The tests shall be conducted at a Government approved test facility (proper safety precautions should be exercised to protect all exposed personnel).

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4.5.23.2.1 Target: Gravel at 200 meters. Half of the ballistic sample (144) shall be equally distributed between three randomly selected launchers and fired to impact at approximately 200 meters from the launcher on a six inch thick loose gravel pad. The gravel pad used shall have a 1/2 inch maximum stone size. The rounds being tested shall be temperature-conditioned at 70° ± 5° F for at least two hours; then fired within 15 minutes after being removed from the temperature soak. The velocity shall be measured at 20 feet from the muzzle of the launcher. If an approved radar system is not used, then the monitoring devices and timer recorders shall be a dual setup, using two sets of screens, each set of screens having an electronic counter. One set of screens shall be placed 14 and 26 feet from the launcher and the other set 15 and 25 feet from the launcher.

NOTE: To ensure that the chronographs are read properly, the ratio of the two times shall not differ from 1.200 by more than ± 0.005. If the ratio differs from 1.200 by more ± 0.005, the velocity results for that round shall be discarded and excluded from the calculation of mean velocity and velocity standard deviation (not more than 8 rounds in a sample size of 144 may be disregarded). If the applicable number of improper ratios is exceeded, additional rounds must be fired. Observation shall be made and results recorded for functioning, mean velocity, and standard deviation. The average of all rounds fired per sample (except above noted improper ratio(s)) shall meet the requirements of 3.5.

4.5.23.2.2 Target: Steel plate at 125 feet. The remaining half of the ballistic sample (144) shall be fired using one launcher to impact at approximately 125 feet from the launcher against a vertical 2 ½ inch thick mild steel plate. If the first 36 cartridges tested successfully penetrate the steel plate, the remaining cartridges may be tested on any thickness steel target (½ inch minimum). With Contracting Officer approval, the plate test samples may be selected prior to sealant cure cycle and tested in increments as the lot is produced for control purpose as well as for lot acceptance.

4.5.23.2.3 Rejection. The lot shall be rejected if:

a. Twelve or more fail high order detonation or fail to defeat 2 ½ inch mild steel plate upon initial impact (see 6.4.4) (major defect).
b. Any primer blows back (as evidenced by perforation of the cup and extrusion of the primer around the firing pin causing difficulty in opening the weapon) (major Defect).
c. Any premature burst occur (see 6.4.3) (critical defect).
d. Mean velocity and/or standard deviation fails to meet the applicable requirement (major defect).
e. Any primer misfires (major defect).
f. Extraction:
    1) Thirty or more cartridge cases cannot be ejected from the launcher by spring operation of the extractor (major defect).
    2) One or more cartridges cases cannot be ejected from the launcher by hand operation of the extractor (major defect).
g. Any projectile sticks in gun bore (critical defect)

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h. Any projectile fails to attain 60% of the mean uncorrected velocity of the lot samples. (A projectile on which a velocity reading was not obtained due to low velocity, shall be considered as a failure) (major defect).
i. Three or more primers show any evidence of gas leakage (smoke deposit or discoloration of base plug or cartridge case base) (major defect).
j. Any metal part separation occurs prior to impact (major defect).

4.5.24 Environmental (informational test only). When required, the following sequence of tests shall be performed as follows:

a. The 72 rounds shall be packaged in their regular shipping containers and subjected to transportation vibration environment in accordance with MIL-STD-331, Test No. B1.
b. After transportation vibration, the rounds shall be unpacked and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. C1.
c. After temperature and humidity cycle, the rounds shall be fired using one launcher, to impact at approximately 125 feet from the launcher against a vertical 2 ½ inch thick mild steel plate. The rounds shall be temperature-conditioned and fired as follows:

24 rounds +165º ± 5º F.
24 rounds ambient (between +40 and +110º F).
24 rounds -65° ± 5° F.

4.5.25 Lot identification. Visually verify that each lot of cartridges is identified as the type, caliber, model and lot number in accordance with MIL-STD-1168.

4.5.26 Heat seal test of packing seal. This test shall be performed in accordance with specification MIL-STD-2073-1. Seals tested shall not be returned to the lot.

4.5.27 Propellant weight. The propellant quantity shall be determined and weighed 100% percent. Any charge that fails to comply with the drawing requirement (taking into account machine accuracy) shall be classified defective and removed from the lot. The accuracy of the machine shall be determined and the prove-out test performed to validate the machine's ability to remove charges outside of the drawing tolerance range. The machine must be capable of detecting and rejecting critical propellant weights, such that the end-product contains less than 1 in a million critical defects (machine error rate not to exceed 1/500 and maximum defect rate not to exceed 0.05%).

4.5.28 Workmanship. Visually verify that all parts and assemblies meet requirements of paragraph 3.9.

5. PACKAGING

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD personnel, or in-house contractor personnel, these personnel need to contact the responsible packaging activity to

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ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cartridge covered by this specification are intended for use in the low-velocity launchers of the M79 and M203 weapon systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number (w/revision level) and date of all reference documents
c. ASSIST Online to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.2.1).
d. Requirements for submission of first article sample.
e. Requirement for submission of design verification sample
f. Certificate of conformance for each lot or shipment of product.
g. Packaging requirements, if other than specified in Section 5.
h. Quality Conformance Inspection, if other than specified in Section 4 of this specification.
i. Provisions for critical characteristic controls.
j. Requirement for submission of inspection equipment designs.
k. Applicable National Stock number (NSN)
l. Serialization requirements, if applicable.
m. Government Furnished Material or Equipment.
n. Lists of drawings, publications and specifications, showing applicable revision dates.
o. Critical inspection equipment requirement (see 6.5).

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.4 Definitions.

6.4.1 Standard deviation. Standard deviation should be calculated by any approved standard deviation formulas.

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6.4.2 Early burst. A projectile that functions in flight further than the arming distance from the weapon muzzle.

6.4.3 Premature burst. A premature burst is considered to have occurred if the projectile functions in the weapon bore or in flight up to the arming distance from the weapon muzzle.

6.4.4 High-order functioning or defeat of mild steel plate. Evidence of high-order functioning should be proper fragmentation of the projectile cup, as determined by the proof director. Evidence of defect of mild steel plate should be through penetration of the plate as determined by the proof director.

6.4.5 Automated acceptance inspection equipment. Equipment for which no, or minimal, human involvement is required in the acceptance determination. This would include equipment employing probes/sensors/transducers that are automatically manipulated to perform measuring and detection functions. If readout is provided, the equipment will be categorized as automated even if operator interpretation is needed in the accept/reject decision.

6.5 Critical Inspection Equipment.

6.5.1 System Reliability. The contractor should provide for each defect identified as Critical, a manufacturing and inspection system that assured no more than one in a million contain the defect. This should be assured by controlling the maximum defect rate produced and the error rate of the inspection equipment such that the product of the two terms is less than one in a million.

6.5.2 Maximum defect rate produced. The maximum defect rate produced should be defined as the largest defect rate expected for the characteristics of concern on a monthly or lot basis. The contractor should establish the maximum defect rate, however, it may not exceed one percent without approval of the procuring contracting officer. Once established, the contractor should monitor the defect rate to assure it does not exceed the maximum rate allotted. If the established maximum defect rate is exceeded, the production of that feature should be terminated until the cause is determined and corrected. Additionally, all products for that period of time should be rejected and reinspected.

6.5.3 Maximum error rate of the inspection system. The error rate of the inspection equipment should be defined as the expected ratio of the number of defective parts accepted to the number of defective parts inspected by the equipment. The contractor, to meet the system reliability as defined in contract, defines the maximum error rate allowed. However, it may not exceed 1/500 without approval of the procuring contracting officer. Based on the maximum error rate defined for the equipment, the contractor should develop demonstration test procedure to demonstrate the error rate of the equipment. The test should be performed using defective parts or reject standards. No part or standard should be accepted during the test. If a part or standard is accepted, the cause for failure should be isolated and corrected and the test rerun. Unless otherwise specified in the detailed item specification, the minimum number of test samples to be run should be equal to 0.7 divided by the error rate (e.g. If the required error rate is 1/1000, the sample size would be 0.7 divided by (1/1000) = 700.)

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6.5.4 Periodic verification. Periodic verification of the system error rate should be performed.

6.5.5 Redundant inspection equipment. The contractor may elect to use redundant inspection equipment in lieu of using a single piece of equipment with a very low error rate. Where redundant equipment is used, the inspection system error rate is defined as the error rates of the multiple pieces of equipment multiplied together.

6.6 Key Word Listing.

Fuze, M550
Grenade Launcher, M203
Escapement, M550

6.7 Revision/Amendments History. The following is a record of changes incorporated into this document

ECP	Description	Date Approved
M5S3008	Delete minor for incorrect color of skirt	06/06/1995
R04Q2021	Initial release of M433 Cartridge MIL-DTL Specification	Aug 2004

6.8 Superseding Documents.

MIL-C-50446A (AR)
22 July 1981
MIL-C-50447A (AR)
10 November 1992
MIL-C-50448A (PA)
14 May 1975
MIL-C-50870C (AR)
22 July 1981
MIL-C-50871C (AR)
22 July 1981
MIL-C-60985A (AR)
20 March 1969
MIL-C-63133 (AR)
14 March 1977

6.9 Interim Revision This interim revision has been prepared at Program Manager request to document in ASSIST Online the version of a draft specification that was placed on contract in 2004. This interim revision has been dated to match the date of the draft specification that was placed on contract. It has been modified slightly to comply with MIL-STD-961 requirements for format, but not content. A fully coordinated revision, compliant with both MIL-STD-961 format and content requirements, will be approved for publication.

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6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Army-AR
(Project 1310-2006-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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